

Decent Coaching Classes
VIII SB
Academic Conquest - III

Subject: Science

Marks: 30

Chapter 5: Inside the Atom

Time: 1 Hour

Instructions:

1. All questions are compulsory unless stated otherwise.
2. Figures to the right indicate full marks.
3. Draw neat labelled diagrams wherever necessary.

Q.1 Fill in the blanks. (Any 5 of 6)

(5 Marks)

- a. According to Dalton's atomic theory, atoms are _____ and _____.
- b. The plum pudding model of atom was put forth by _____.
- c. Rutherford put forth his nuclear model of the atom in the year _____.
- d. The maximum number of electrons that a shell can accommodate is given by the formula _____.
- e. Atoms of the same element having the same atomic number but different mass numbers are called _____.
- f. _____ is used as a moderator in a nuclear reactor to slow down neutrons.

Q.2 Answer in one sentence. (Any 5 of 6)

(5 Marks)

- a. What is an atom?
- b. Which scientist demonstrated that the negatively charged particle in an atom has a mass 1800 times less than a hydrogen atom?
- c. What are valence electrons?
- d. State any two uses of isotopes.
- e. What is meant by the atomic number of an element?
- f. Name the two types of nucleons present in the nucleus of an atom.

Q.3 Match the pairs.

(4 Marks)

Group 'A'	Group 'B'
a. Proton	i. Electrically neutral particle
b. Electron	ii. Positively charged particle
c. Neutron	iii. Same atomic number, different mass number
d. Isotope	iv. Negatively charged particle

Q.4 Give scientific reasons. (Any 3 of 4)

(6 Marks)

- a. Almost the entire mass of an atom is concentrated in its nucleus.
- b. An atom is electrically neutral in spite of containing negatively charged electrons.
- c. The atomic mass number of an element is always a whole number.
- d. Rutherford's atomic model was considered to be unstable according to an established law of physics.

Q.5 Distinguish between. (Any 2 of 3)

(4 Marks)

- a. Thomson's atomic model and Rutherford's atomic model
- b. Proton and Neutron
- c. Nucleus and extranuclear part of an atom

Q.6 Answer the following. (Any 2 of 3)

(6 Marks)

- a. Explain the important postulates of Bohr's stable orbit atomic model.
- b. State the relationship between the valency of an element and the number of its valence electrons.
- c. An atom has atomic number 12 and mass number 24. Deduce the number of protons, electrons and neutrons in it, and write its electronic configuration.